

Operation Procedure:

Connect the Charger to Battery first using the Battery Connector cable. Apply mains power using a 16A rated mains socket. The charger will blink all five indications once and a characteristic beep sound at power-up, the Green mains indication will become steady and the charger will start charging. The Battery Bar Indications will start animating up-down with current increasing gradually. Ensure the charger is kept in a well-ventilated area and the ventilation slots on the charger must be un-obstructed for air flow. As the charging current increases, the charger turns on the internal cooling fan and turns off the fan once the current goes down or internal temperature falls. Once the Battery voltage reaches set voltage and current falls to a pre-defined level, the Battery Bar Indications becomes steady. The battery is almost full charged at this instant and can be disconnected and used for backup. However, the charger continues to charge the battery for some more time till the battery draws less than 5% of the CC current. Once the battery is fully charged the charger stops and gives a characteristic beep sound will the Battery Bar indication steady on.

LED indications:

Indication	Condition
Mains Led Steady on	Charging On from Mains
Mains Led Blink	Mains out of Acceptable range.
Battery Bar All Four Led Steady on	Charging Complete
Battery Bar One or More Led Up-Down	Charging On
One or More Battery Bar Led Steady on	Battery Voltage Level showing – Not Charging
1 Battery Bar Led 25 Blink	Deep Discharged Battery
2 Battery Bar LED 100,25 Blink	Fault Indication occurring immediately after turning on power.
	Battery Reverse Connection OR Output Short OR Battery Voltage too High.
	Battery Connection Open Circuit OR PCM Disconnect in already full charged battery.
2 Battery Bar LED 100, 75 Blink	Fault Indication occurring 10-15 Sec. after Power On
	PCM Disconnect due to battery full charge.
2 Battery Bar LED 100, 75 Blink	Over Temperature Fault
2 Battery Bar LED 50, 25 Blink	Internal Fault–No output Current
All Four Battery Bar LED Blink	Internal Fault –Over Current
3 Battery Bar LED 100, 75, 50 Animated Blink	CAN Communication Failure (For CAN Models)

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ECOSTAR®

E-Vehicle Lithium Battery Charger 1000W

AIS 156 II
COMPLIED



DESIGNED & MADE
INDIA

This 1000W Lithium Battery Charger is a SMPS type battery charger optimized for charging LFP or NCM or NCA chemistry Li-Ion battery packs for Light E-Vehicle application. Various models are available in this series for charging 13-Cell-Series to 24-Cell-Series Lithium Battery packs.

Standard Features

- Charging Current and Voltage profile optimized for LFP or NCM or NCA chemistry battery packs for Light E-Vehicle application. Can be customized for other Li-Ion chemistry such as LTO also.
- Intelligent algorithm to detect battery BMS disconnection and auto-adjustment of CC-CV settings to maximize charge return and enhance cell equalization.
- High Efficiency of more than 90% - lower electricity consumption and less heat generation.
- Active Power Factor Correction to reduce electricity consumption.
- High charging current even at low mains voltages for faster charging.
- Electronic Reverse Battery Protection - enhanced safety and avoid inconvenient fuse replacement in case of accidental reverse connection.
- Pre-Charge and Soft-Start Function to safely recharge deep discharged batteries.
- Enhanced safety with Electronic Reverse Battery Protection.
- Informative Battery Bar Graph and Mains indication along with audio alarms.
- Compact size for easy handling.
- Rugged metal enclosure.

Additional Features with CAN Interface

- Features are compliant with the requirements of AIS156 Phase2 amendments.
- Integrated CAN Communication Interface for Battery Pack Identification.
- Charger CV CC Parameters configurable through CAN Interface.

Model List :

Models – Low Voltage	ES5517	ES5616	ES5716	ES5816	ES58516
Battery Pack Type	15S-LFP 13S-NCM	15S-LFP 13S-NCM	16S-LFP	16S-LFP 14S-NCM	14S-NCM
Nominal Battery Voltage	48V	48V	51V	51V	51V
Max. Charging Voltage	55.0V	56.0V	57.0V	58.0V	58.5V
Max. Charging Current	17A	16A	16A	16A	16A
Models – High Voltage	ES6712	ES6912	ES7212	ES83510	
Battery Pack Type	19S-LFP 16S-NCM	19S-LFP 20S-NCM	20S-LFP 17S-NCM	24S-LFP 20S-NCM	
Nominal Battery Voltage	60V	60V	60V	72V	
Max. Charging Voltage	67.0V	69.0V	71.0V	84.0V	
Max. Charging Current	12A	12A	12A	10A	

* Other Models are also available based on requirement.

* For products with CAN communication interface the charger will automatically set max charging voltage and charging current as communicated by BMS through CAN command.

Specifications

Max Output Power	1000W MAX	
CV Voltage Range	53V-84V	Configurable through CAN Communication Interface or factory preset .
CC Current Range	5A-17A	
CC Current Tolerance	CC Setting $\pm$ 0.5A	
CV Voltage Tolerance	CVSetting $\pm$ 0.3V	
Efficiency	91% Typical	
Input Power Factor	Active PF correction, PF>0.98, ITHD < 5%	
Mains Opreating Range	115VAC -280VAC $\pm$ 10V	
Mains Opreating Frequency	40 Hz TO 60 Hz	
Max Input Current	4.5A (Output current de-rated at low mains to limit max Input current)	
Cooling System	Forced Air Cooled	
Visual Indication	LED Based	
Audio Indications (Buzzer)	Charger Startup, Battery Charge Completion – Characteristic Pattern Beep. Mains Failure or any Fault – Single Beeps	
Protection	Reverse Battery (Electronically) Mains High Voltage (Withstand up to 320VAC RMS.), Mains Over Current (by Fuse) Thermal Protection (Electronically)	
Soft Start Rate	Current Rise rate 0.9A/Sec typical. (0 to 17A in 17 Seconds).	
Charge Termination	1. CV Mode Current < 3A for 15Minutes or 2. CV Mode Time Exceeds 1Hr 50Minutes or 3. Total Charging Time Exceeds 7 Hours	
Pre-Charge Setting	If Battery Pack Voltage below 75% of CAN Configured CV Setting, Current Limited to CC/4. If No CAN, below 40V Current Limited to CC/4. Below 33V current limited to 2.0A.	
CAN Interface (Optional)	ISO11898 CAN 2.0B Compliant Physical Interface. Software Protocol customizable as per BMS requirement specification.	
Earth Leakage Protection (Optional)	Inline ELCB Device compliant as per class I of IS 12640-1	
Charger Operating Temprature	0°C TO 45°C	
Humidity	95% RH Non-Condensing	
Battery Cable	1.3m, 4sqmm, Insulated 2 Core Cable With Anderson SB50 Or SB75x for equivalent	
Mains Cable	1.8m, 1sqmm 3core ISI Marked Cable With 6A Plug	
Enclosure	Powder coated sheet metal cabinet	
Dimension / Net Weight	290mm x 135mm x 105mm / 3.2kg	